ALICE L WALTON SCHOOL OF MEDICINE

History-Taking Approach to Anemia

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Learning Objectives (2)

After completing this brick, you will be able to:

1. Identify the two definitions of anemia: physiological and clinical.
2. Predict the cause of anemia through six key history questions.

CASE CONNECTION

A 28-year-old woman presents to your clinic complaining of fatigue. On examination she looks pale, but the physical exam is otherwise unremarkable. A third-year medical student is rotating with you and remarks, "Dr. S, this patient looks anemic. How do I figure out what's causing it?" Consider your answer to this medical student as you read, and we'll revisit it throughout the brick.

GO TO CONCLUSION

"Anemia" is not a specific diagnosis- it is an abnormality which indicates the presence of an underlying disease. Simply chasing the anemia may cause you to miss a significant underlying cause that warrants further treatment. The most important part in the evaluation of an anemic patient is the **history.** To embark on this discovery, let's begin with six key history questions:

- Is the patient truly anemic?
- Is there evidence of blood loss?
- Is the anemia inherited or acquired?
- Is there evidence of hemolysis?
- Has the patient been exposed to medication or toxins that can result in anemia?
- Is there evidence of a systemic illness or non-hematologic organ dysfunction?

We will explore each of these questions separately to demonstrate their value in the diagnosis of anemia.

Is the patient truly anemic (LO #1,2)?

There are two definitions of anemia. The **physiologic definition** is an insufficient oxygen-carrying capacity in the blood to meet the oxygen requirement. The **clinical definition** is decreased hemoglobin (or hematocrit) compared to the reference population. Please note, normal hemoglobin and hematocrit is a range, derived from the mean of a normal reference population +/- 2 standard deviations.

Table 1

RBC index	Definition	Typical reference range
Red blood cell (RBC) count	Number of RBCs in the blood	Males: 4.3-5.9 × 10^{12}/L Females: 3.5-5.5 × 10^{12}/L
Hemoglobin (Hgb)	Concentration of Hgb in the blood	Males: 14-18 g/dL Females: 12-16 g/dL
Hematocrit (Hct)	Percentage of blood composed of RBCs	Males: 40%-54% Females: 36%-48%
Mean cell volume (MCV)	Average volume (size) of RBCs	80-100 fL
Mean cell hemoglobin (MCH)	Average weight of hemoglobin per RBC	27-31 pg/cell
Mean cell hemoglobin concentration (MCHC)	Average concentration of Hgb per RBC	32 to 36 g/dL
Red blood cell distribution width (RDW)	Variability of the MCV	12% to 15%

Table 1

It is important to compare the hemoglobin value to the appropriate reference population. Let's make … and sex.

- Age: Children have a different "normal" range than adults from birth until puberty; the hemoglobin approaches the adult range after puberty. The "normal" hemoglobin varies at different ages during childhood, so you must use an age-matched reference range. The laboratory should provide you an age-matched reference range as part of the complete blood count (CBC).

CLINICAL CORRELATION

Infants usually experience a physiologic anemia around 8-12 weeks of age, with a nadir that can reach 9.5-10 g/dL. After birth, RBC mass normally declines in response to an increase in the availability of oxygen and downregulation of erythropoietin. RBC count decreases until oxygen delivery is inadequate for metabolic demand and erythropoietin production is stimulated again. Pre-term infants experience a more abrupt and profound decline in their hemoglobin.

- Sex: Men tend to have a higher hemoglobin than women, probably because testosterone has an erythropoietic effect. Again, the laboratory should include appropriate reference ranges in the CBC report.

Hemoglobin is measured as a ratio: grams of hemoglobin/deciliter of blood (g/dL). Blood is composed of both red cells and plasma. If there is a change in the plasma volume (denominator) without changing the grams of hemoglobin, the overall ratio appears decreased. This can occur in situations in which there is an increased plasma volume, such as pregnancy, congestive heart failure, splenomegaly, or rece… on is termed **"dilutional anemia"** …nt and transfusion without recognizing this possibility. So, when you have an anemic patient, take a step back and ask "is the patient truly anemic?"

CLINICAL CORRELATION

A 16-year-old male presented to the ER with a 2-week history of diarrhea and 1-week history of nausea/vomiting. He was admitted for sepsis, hypovolemic shock, acute kidney injury/failure, and lactic acidosis. His admission CBC showed a hemoglobin of 15.3 g/dL (normal: 14-18 g/dL), with a repeat CBC (less than 24 hours later) showing a hemoglobin of 9.9 g/dL. Of note, his other CBC values (white blood cell count, platelet count) decreased as well.

Patients admitted to the ER, particularly those that are dehydrated such as this patient, are given large boluses of IV fluids. This can dilute out the red cells, showing a dilutional anemia. A helpful clue that this drop in hemoglobin was dilutional is the fact that the white blood cell count and platelet count dropped as well.

CLINICAL CORRELATION

Multiple physiologic changes can occur to the hematopoietic system during a "normal" pregnancy, one of which is the increase in plasma volume relative to the increase in RBC mass that can be associated with a modest decrease in hemoglobin concentration, referred to as physiologic or dilutional anemia of pregnancy. Typically, anemia in pregnancy is defined as <11 g/dL in the first and third trimesters and …

How do we know if the … "normal", physiologic changes? If hemoglobin levels are lower or higher than expected or the anemia is associated with additional symptoms or evidence of iron deficiency, additional evaluation is warranted.

Is there evidence of blood loss (LO #2)?

Blood loss is a common cause of anemia. Therefore, it is important to ask thorough, specific questions to identify its source.

- **Gastrointestinal System History:**
  - Has there been any change in bowel habits?
  - Has the patient noticed any blood in their stool or dark, tarry stools (called melena)?
  - Does the patient have "heartburn" or symptoms of gastroesophageal reflux?
- **Reproductive/Menstrual History:**
  - Menstrual History: How long do the patient's periods last? How many pads or tampons do they use per period? Do they pass clots during their period?
    - Menstrual blood loss is a very common cause of iron deficiency anemia. It is important to recognize that each patient's period is relative to them; one person's light period is one person's heavy period. Asking specific questions about number of pads and length of period can assess the risk of iron loss.
  - Reproductive History: How many pregnancies has the patient had and how far apart…
    - Each preg… **milligrams of iron lost** (a huge of amount of iron to replace). Women who have had multiple pregnancies, especially at short intervals, are highly likely to become iron deficient.
- **Urinary System History:**
  - Has there been any evidence of blood in their urine **(hematuria)**? This is an uncommon but occasional source of blood loss.
- **Frequent phlebotomy (blood draws) or blood donation:** Each unit of blood represents about 200 mg of iron loss and can be a source of iron deficiency in the absence of supplementation.

What is a common source of blood loss in women of child-bearing age?

Is the anemia inherited or acquired (LO #2)?

There are certain imp… rited cause of the patient's anemia.

- Family history: Most patients will not know exact diagnoses of family members. However, asking if someone in their family had to receive treatment for anemia (iron supplementation, blood transfusion) may yield a positive result.
- Age at onset: Typically, inherited causes of anemia present at an earlier age.
- Treatment for anemia: Ask if the patient ever received treatment for anemia, such as iron supplementation.
- Transfusion History: If the patient had transfusions at a young age or transfusions without an obvious cause, such as hemorrhage or surgery, that may suggest an inherited process.
- Ethnic background: Some inherited causes of anemia are more prevalent in certain areas of the world (typically arising as a resistance to malaria).

Is there evidence of hemolysis (LO #2)?

Hemolysis refers to premature destruction of red blood cells and can cause a few characteristic symptoms.

- Jaundice (yellowing of the skin and mucous membranes): Bilirubin is a breakdown product of heme. The bilirubin, specifically unconjugated or indirect bilirubin, increases in hemolysis due to increased red cell turnover, leading to jaundice.
- Dark urine: This can occur due to hemoglobinuria (hemoglobin in the urine), typically if the hemolysis is intravascular.
- Abdominal discomfort due to splenomegaly: The spleen is the site of hemolysis in ma… re, splenomegaly (enlarged spleen) can be a consequence of chronic hemolysis.

Has the patient been exposed to medication or toxins that can result in anemia (LO #2)?

Typically, the answer to this question is yes! Many, many medications can be associated with anemia. A complete drug history, including over the counter and dietary supplements, is essential in the evaluation of anemia. The following are just a few examples (this list is by no means complete):

1. Agents associated with marrow aplasia/marrow suppression: Chemotherapy drugs, insecticides, anticonvulsants such as phenytoin.
2. Agents that can cause GI blood loss: Non-steroidal anti-inflammatories (NSAIDS, like ibuprofen).
3. Agents associated with hemolytic anemia: oxidant drugs such as dapsone, sulfa drugs, primaquine (used to treat malaria).
4. Agents causing maturation defects: folate antagonists such as trimethoprim (antibiotic) and methotrexate (used as chemotherapy and to treat autoimmune diseases).

CLINICAL CORRELATION

Patients with glucose-6-phosphate dehydrogenase deficiency (G6PD deficiency) are missing an enzyme necessary to detoxify hydrogen peroxide (a reactive oxygen species), making the red cells susceptible to oxidant stress. A majority of patients are asymptomatic until presented with an oxidant stressor, such as medications (primaquine, dapsone … The classic presentation is a previously healthy … ressor such as a medication or infection, presents with signs of hemolytic anemia (fatigue, pallor, jaundice).

Is there a systemic illness or non-hematologic organ dysfunction (LO #2)?

Again, the answer is typically yes! Many underlying conditions, such as infection, chronic inflammation, renal disease, liver disease, endocrine disease, and malignancy, can be associated with anemia.

CASE CONNECTION

BACK TO INTRODUCTION

After working through the key history-taking questions with your student, he asks the patient these additional questions and reports the following answers:

"The patient is not currently pregnant and has no history of congestive heart failure or splenomegaly. She reports heavy periods that last for 7 days, using 5 pads/day, and she occasionally passes blood clots. She has four children ranging between 2 and 6 years of age. She eats a regular diet, with meat. She has not noticed any change in bowel habits, dark stools, or heart burn. She has no family history of anemia/blood transfusions but was told she was anemic during a previous pregnancy. She has not experienced any jaundice or dark urine. She takes ibuprofen for occasional menstrual migraines but no other relevant medications. She has no other significant past medical history."

After complimenting your student on his thorough history-taking skills, you summarize these findings.

"Based on what the p… like she has quite significant menstrual blood loss. Additionally, given that she had many children close together and particularly as she was diagnosed with anemia during a previous pregnancy, it is likely that her body was not given enough time to replenish the iron she lost. Overall, I'm concerned that she has iron deficiency anemia due to the extensive menstrual/pregnancy iron loss. Let's confirm that with some additional laboratory testing."

Summary

- Anemia is not a specific diagnosis but an indication of an underlying disease, and it's crucial to identify the root cause rather than just treating the anemia.
- The evaluation of an anemic patient should include a thorough history check, investigating potential blood loss, inherited or acquired anemia, evidence of hemolysis, exposure to medication or toxins that can result in anemia, and evidence of a systemic illness or non-hematologic organ dysfunction.
- There are both physiological (insufficient oxygen-carrying capacity in the blood) and clinical (decreased hemoglobin compared to the reference population) definitions of anemia.
- Blood loss, through the gastrointestinal system, menstruation, pregnancies, or frequent blood draws, is a common cause of anemia.
- Inherited anemia can be identified through family history, age at onset, treatment for anemia, transfusion history, and ethnic background.
- Many medications and underlying conditions such as infection, chronic inflammation, ren… disease, and malignancy can be … ated with anemia.

Review Questions

1. Which of the following laboratory parameters is used to establish a clinical diagnosis of anemia?

A. ○ Direct antiglobulin test (DAT)
B. ● Hemoglobin
C. ○ Bilirubin, indirect/unconjugated
D. ○ Mean corpuscular volume (MCV)
E. ○ Red cell distribution width (RDW)

Hide Explanation

The correct answer is B) hemoglobin. The clinical definition of anemia is decreased hemoglobin (or hematocrit) compared to a normal reference population. DAT (answer choice A) is a measure of autoimmune hemolysis, as it detects antibodies on the red cell surface. Indirect bilirubin (answer choice C) is an overall measure of hemolysis. MCV (answer choice D) is a measure of the size of the red cells, an important parameter we use to narrow the differential once we have an established diagnosis of anemia. RDW (answer choice E) is a measure of the variability in the size of the red cells, an important parameter we use to narrow the differential once we have an established diagnosis of anemia.

2. A 7-year-old child presents to the clinic for a follow-up. His mother mentions he receives red blood cell (RBC) transfusions once per month for treatment of his anemia, with similar treatments being required for his aunt and cousin. Which key history question is displayed in this clinical vignette?

A. ○ Is there evidence of blood loss?
B. ○ Is the patient truly anemic?
C. ○ Is there evidence of hemolysis?
D. ● Is the anemia inherited or acquired?
E. ○ Has the patient been exposed to medication or toxins that can result in anemia?

Hide Explanation

The correct answer is D, is the anemia inherited or acquired? The patient is young, receives multiple transfusions to treat his anemia, and has a positive family history, suggesting an inherited cause of his anemia. There are no identified sources of blood loss mentioned in the vignette to suggest answer choice A. Answer choice B is incorrect, as the patient has no identified conditions associated with an increased plasma volume and definitely has anemia as he requires multiple transfusions for treatment. The patient has no evidence to suggest hemolysis, such as jaundice or dark urine (answer choice C). Answer choice E is incorrect, as the clinical vignette mentions no relevant

How would you rate this Brick?

References

- Kumar V, Abbas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease.* 10th ed. Elsevier; 2021.
- Hoffman R, Benz EJ, Silberstein L, et al. *Hematology: Basic Principles and Practice.* 8th ed. Elsevier; 2023.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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Objective Confidence: 2/2